

Computer Science & Information Technology

Discrete Mathematics

DPP: 1

Graph Theory

- Q1** How many simple graphs are possible with 5 vertices and 3 edges?
- Q2** A non-directed graph contains 16 edges and all vertices are of degree 2. Then number of vertices in G is _____.
- Q3** G is an undirected graph with n vertices and 25 edges such that each vertex of G has degree at most 4. Then minimum possible value of n is _____.
- Q4** G is an undirected graph with n vertices and 35 edges such that each vertex of G has degree at least 3. Then maximum possible value of n is _____.
- Q5** If G be a graph with 5 vertices and 7 edges, then
 (A) $\delta > (G) 2$
 (B) $\Delta > (G) 4$
 (C) $\delta \leq (G) 2$
 (D) $\Delta \leq (G) 2$
- Q6** Find the number of edges of an undirected graph having degree sequence 2, 4, 4, 3, 4, 1?
- Q7** An ordered n -tuple $(d_1, d_2, d_3, \dots, d_n)$ with $d_1 \geq d_2 \geq d_3 \dots \geq d_n$ is called graphic if there exists a simple undirected graph with n vertices having degrees $d_1, d_2, d_3, \dots, d_n$ respectively.
- Which of the following is a graphic sequence?
 (A) 5, 3, 3, 2, 2, 1
 (B) 2, 1, 1, 1, 1, 1
 (C) 6, 5, 4, 3, 2, 1
 (D) 5, 5, 2, 2, 1, 1
- Q8** There are 24 routers in an organization. Find the number of cables required to connect them such that each router is connected with exactly 6 other routers.
- Q9** Consider a complete graph with size 2016. Suppose after deletion of 2 vertices from the above graph, the modified graph have x number of edges and y number of vertices. Find the value of $x - y$?
- Q10** The degree sequences of a simple graph is the sequence of the degree of the nodes in the graph in decreasing order. Which of the following sequences cannot be the degree sequence of any graph?
 I. 7, 6, 5, 4, 4, 3, 2, 1
 II. 6, 6, 6, 6, 3, 3, 2, 2
 III. 7, 6, 6, 4, 4, 3, 2, 2
 IV. 8, 7, 7, 6, 4, 2, 1, 1
 (A) I and II
 (B) III and IV
 (C) IV only
 (D) II and IV

Answer Key

- Q1** 120~120
Q2 16~16
Q3 13~13
Q4 23~23
Q5 (C)
Q6 9~9
Q7 (A)
Q8 72~72
Q9 1829~1829
Q10 (D)


[Android App](#)
[iOS App](#)
[PW Website](#)

Hints & Solutions

Q1 Text Solution:

Total number of edges possible with 5 vertices
 $= \frac{5 \times 4}{2} = 10$

No. of simple graphs with 3 edges = ${}^{10}C_3 = 120$

Q2 Text Solution:

$\frac{n \times \text{degree}}{2} = \text{No. of edges}$
 $\Rightarrow \frac{n \times 2}{2} = 16$
 $\Rightarrow n = 16$

Q3 Text Solution:

$\frac{2E}{V} \leq \Delta$
 $\Rightarrow \frac{2 \times 25}{V} \leq 4$
 $\Rightarrow V \geq 12.5$
 $\Rightarrow V = 13$

Q4 Text Solution:

$\delta \leq \frac{2E}{V}$
 $\Rightarrow 3 \leq \frac{2 \times 35}{V}$
 $\Rightarrow V \leq 23.33$
 $\Rightarrow V = 23$

Q5 Text Solution:

$\delta \leq \frac{2E}{V} \leq \Delta$
 $\Rightarrow \delta \leq \frac{2 \times 7}{5} \leq \Delta$
 $\Rightarrow \delta \leq 2.8 \leq \Delta$
 $\Rightarrow \delta \leq 2, \Delta = 3$

Q6 Text Solution:

Total degree = $2 + 4 + 4 + 3 + 4 + 1 = 18$
 No. of edges = $\frac{18}{2} = 9$

Q7 Text Solution:

(A) 5,3,3,2,2,1
 $\Rightarrow 2,2,1,1,0$
 $\Rightarrow 1,0,1,0$
 $\Rightarrow 1,1,0,0$
 $\Rightarrow 0$ (Graphic Sequence)
 Option B and C are invalid as the number of odd degree vertices is odd.
 (D) 5,5,2,2,1,1

$\Rightarrow 4,1,1$ (Invalid)

Q8 Text Solution:

Number of cables = $\frac{24 \times 6}{2} = 72$

Q9 Text Solution:

Let there be n vertices.

So, ${}^nC_2 = 2016$

$\Rightarrow n = 64$

Degree of each vertex = $64 - 1 = 63$

After deletion of two vertices,

No. of edges (x) = $2016 - 63 - 63 + 1 = 1891$

No. of vertices (y) = $64 - 2 = 62$

$x - y = 1891 - 62 = 1829$

Q10 Text Solution:

(A) 7,6,5,4,4,3,2,1

$\Rightarrow 7,6,5,4,4,3,2,1$

$\Rightarrow 3,2,2,1,0$

$\Rightarrow 1,1,0$

$\Rightarrow 0$ (valid)

(B) 6,6,6,6,3,3,2,2

$\Rightarrow 5,5,5,2,2,1,2$

Rearrange

$\Rightarrow 5,5,5,2,2,1,2$

$\Rightarrow 4,4,1,1,1,1$

$\Rightarrow 3,0,0,0,1$

Rearrange

$\Rightarrow 3,1,0,0,0$ (Invalid)

(C) 7,6,6,4,4,3,2,2

$\Rightarrow 5,5,3,3,2,1,1$

$\Rightarrow 4,2,2,1,0,1$

Rearrange

$\Rightarrow 4,2,2,1,1,0$

$\Rightarrow 1,1,0,0,0$

$\Rightarrow 0$ (valid)

(D) 8,7,7,6,4,2,1,1

Total 8 vertices are there so max degree can be 7, so invalid.



[Android App](#)

| [iOS App](#)

| [PW Website](#)